



ARM946E_88

TSMC 0.18G

(Rev 1)

Core Configuration Report

ARM946E_88 Core Configuration Report

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Change history

Issue	Date	Change
0.1	24 June, 2002	First draft.

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1 ABOUT THIS DOCUMENT

1.1 References

This document refers to the following documents.

Ref	Doc No	Title
1.	ARM DDI 0201	ARM946E-S (Rev.1) Technical Reference Manual
2.		

1.2 Terms and abbreviations

This document uses the following terms and abbreviations.

Term	Meaning
I-Cache	Instruction Cache
D-Cache	Data Cache
TCM	Tightly Coupled Memory
I-TCM	Instruction TCM
D-TCM	Data TCM
RAM	Random Access Memory
P&R	Place and Route
BIST	Built-In Self Test

2 SCOPE

The purpose of this document is to provide a summary of the design configuration for a core hardening of the ARM946E-S Rev.1.1 (r1) macrocell.

Configuration is performed as part of the ARM946E-S CPU core hardening flow. The document details the physical aspects of the configuration with respect to RAM generation and also the soft configurable items such as cache and memory sizes and 'design for test' options.

3 INTRODUCTION

The ARM946E-S is a synthesisable macrocell combining the ARM9E-S processor core with a configurable memory system. The macrocell is designed to be configurable for a wide range of applications. The ARM946E_88 represents a specific configuration of the ARM946E-S chosen for the foundry program and a configuration report is detailed in the following sections.

4 SUMMARY

4.1 General

Silicon vendor	Foundry: TSMC
Process name/geometry	TSMC 0.18um CL018 (baseline process) 1P6M,1.8V/3.3V
Polysilicon/Metal	1P6M available, 1P5M used.

Process condition	Temperature	Voltage	Process
Slow (worst case)	+125 degrees C	1.62V	1
Typical	+25 degrees C	1.80V	1
Fast (best case)	0 degrees C	1.98V	1

Standard cell library	
Vendor	Artisan TSMC 0.18um SAGE-X
Version	2001q3v1

Integrated clock gate cell library	
Vendor	Artisan TSMC 0.18um SAGE-X
Version	2001q3v1

Compiled RAM library	
Vendor	Artisan HS-SRAM-SP SRAM Generator, TSMC 0.18u Process
Version	2000Q3V1

4.2 ARM946E configuration

Module name	ARM946E_88
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Memory	Implementation	Size
Instruction Cache (I-Cache)	Compiled RAM	8 KBytes
Data Cache (D-Cache)	Compiled RAM	8 KBytes
Instruction Tag (I-Tag)	Compiled RAM	4 instances of 64 x 21-bit
Data Tag (D-Tag)	Compiled RAM	4 instances of 64 x 21-bit
Instruction TCM (I-TCM)	N/A	Not present in macrocell, but TCM interface exposed.
Data TCM (D-TCM)	N/A	Not present in macrocell, but TCM interface exposed.

Note The “D-Dirty” Memory (1 x 64 x 8-bit) used by the Data cache write-back mode, is implemented as flip-flops (not compiled RAM). Manufacturing test is performed using scan testing.

Configuration parameter setting	
BIST present?	Yes, BIST is enabled.
BIST algorithm	Default ARM946ES BIST implementation using generic 6N algorithm.
Number of BIST controllers	6
Test wrapper present	No. The test wrapper (A946ESWrapper.v) must be added externally.
Clock gating implemented?	Yes.

Synthesis and scan test/ATPG	
Number of scan chains	4

Note In addition to the number of scan chains above, an external test wrapper scan chain is also required to test the internal and external macrocell logic test shadow.



5 MEMORY CONFIGURATION

5.1 Memory requirements

The ARM946E-S CPU Core allows for a configurable memory system. For an optimal hardening solution the cache memories are fixed in size. Each memory has an instruction and data component, the size of which can be independently selected:

5.2 Physical configuration

The following gives a brief outline of how the cache is implemented in the ARM946E_88 and how this relates to the physical size of the generated RAM's. For a more detailed description of the cache architecture, please refer to the ARM946E-S Technical Reference Manual [Ref.1].

5.2.1 Cache sizes

The I-Cache and D-Cache are four-way set associative. This means that the caches are split up into four separate memory blocks making up the four sets.

The D-Cache supports write-back mode which means that the D-Cache may have to be written to and read from at the same time. Because of this each D-Cache set is split further into two physical memory blocks. This makes the two single-port RAM's appear as a one dual-port RAM. Therefore, in the ARM946E_88 implementation there are 8 physical memory blocks making up the D-Cache and 4 memory blocks making up the I-Cache.

For example, to implement 8 KB of data and instruction cache:

Data Cache -	256 x 32 instantiated eight times ¹ .
Instruction Cache -	512 x 32 instantiated four times.

5.2.2 Tag sizes

Each cache segment has an associated TAG RAM for storing the cache line address. The size of the address TAG is dependent upon the cache sizes and is therefore fixed dependant on the cache size. There are four physical memory blocks for each TAG.

For the 8 KB caches above this yields:

Data TAG -	64 x 21 instantiated four times.
Instruction TAG -	64 x 21 instantiated four times.

5.2.3 Data Cache write back memory

There is one further memory block in the ARM946E_88 macrocell. This is the D-Dirty Memory and it stores the dirty bits during data cache write-back mode.

For the 8 KB D-Cache configuration, the size of the dirty bits memory is 64 x 8-bit. This is implemented as flip-flops. Manufacturing test is performed using scan testing.

¹ As the data cache is implemented as a single 32-bit wide block, it has a byte write capability

5.2.4 Tightly-Coupled Memory

The I-TCM and D-TCM are both external to the ARM946E_88 macrocell, and the TCM interfaces are exposed so that these memories can be added during the system chip integration task.

6 SOFT CONFIGURATION

6.1 Design for test

The ARM946E_88 hardened macrocell is fully scan inserted to allow testing of the standard logic. In addition, the macrocell contains BIST hardware to test the RAM's.

6.1.1 Scan insertion

Scan insertion is carried out during the synthesis process using Synopsys test insertion tools. Scan test is implemented using a multiplexed flip-flop style. The number of scan chains was configured at the synthesis stage. The standard synthesis process balances the scan chain lengths. The scan insertion process will create top-level ports for the scan data – SCANIN1 to SCANIN<n> and SCANOUT1 to SCANOUT<n> where n is the number of chains. The ARM946E_88 scan enable pin is SCANEN.

6.1.2 BIST controllers

The ARM946E_88 has been configured to utilise six separate BIST controllers, one for each cache, TAG and external TCM memory. The BIST controllers are part of the ARM946E_88 macrocell. This configuration allows for the full BIST functionality described in the ARM946E-S Technical Reference Manual [Ref.1].

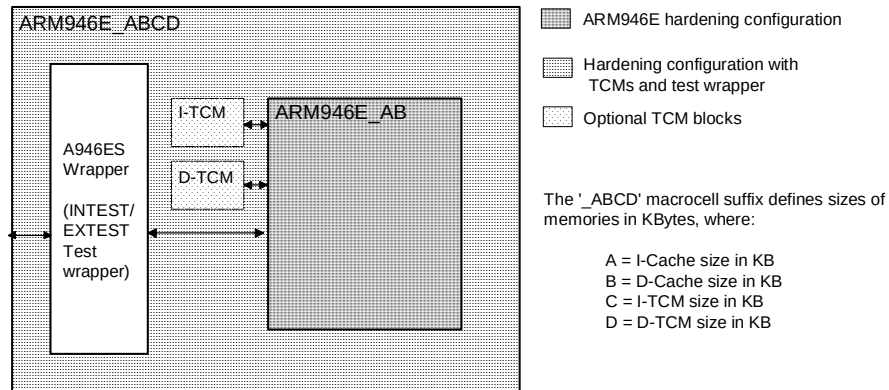
BIST can be activated by scan means using the test wrapper (INTEST/EXTEST) around the hardened macrocell—see section 7 for further details of the test wrapper.



7 HIERARCHY OF THE HARDENED DESIGN

7.1 The hardened level

Below is a diagram showing the ARM946E_88 hardened macrocell within a typical hierarchical level that adds the test wrapper and optionally adds Tightly Coupled Memories (TCMs):



The above figure indicates the two levels at which the hardening activity can be performed. The ARM946E_AB level contains the ARM9E-S core and the instruction and data caches.

The ARM946E_ABCD level includes the test wrapper and the tightly coupled memories (TCM's) in addition to the ARM946E_88. The 'ABCD' notation indicates the size of the caches and TCM's as indicated in the figure above.

In this configuration, A=8 and B=8 since I-Cache and D-Cache are respectively configured for 8KB in size.

A Design Simulation Model (DSM) can be implemented at either of these levels. A DSM is a timing accurate simulation model targeted at a variety of simulators that allows faster simulation than pure gate level netlist simulations.

For the ARM946E_88, a DSM is provided for the ARM946E_AB hierarchical level.

7.2 Test wrapper

The ARM946E_ABCD illustrates inclusion of the test wrapper (A946ESWrapper) which provides an extra scan chain that can be used in either an 'INTEST' or 'EXTEST' test mode. The INTEST mode provides a scan source to input pins on the ARM946E_88 and a capture cell on every output. The INTEST scan chain can be used to test the test shadow logic between each ARM946E_ABCD input pin and any macrocell register it connects to, and also from any macrocell registers which have a logic path to an output pin.

It is also possible to put the test wrapper into EXTEST mode. This allows the System-on-Chip integrator to generate ATPG vectors which test the shadow logic between the ARM946E_88 macrocell and flip-flops outside the macrocell.

The test wrapper can also be used as an additional method of activating the RAM BIST functionality during manufacturing test.

For further information on the test wrapper please refer to the ARM946E-S Technical Reference Manual and the ARM946E_88 Core Design Flow Guide.